

Synthetic CT for MRI-based Pelvic Radiotherapy

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Abstract. This study compares deep learning models for MRI-to-CT translation in radiotherapy planning. An Autoencoder is evaluated, using a supervised encoder-decoder structure, alongside Pix2Pix, which employs adversarial learning for enhanced realism. Metrics like MAE, MSE, PSNR, SSIM, and SNR show that Pix2Pix offers better reconstruction accuracy and sharper images, making it more suitable for radiotherapy despite the Autoencoder's slightly better structural preservation. These findings emphasize the benefits of GAN-based approaches for medical image synthesis.

1 Introduction

Accurate imaging is vital in radiotherapy planning to ensure the treatment is delivered effectively. [1]. Traditionally, computed tomography (CT) has been the standard imaging method due to its ability to provide electron density information essential for dose calculations [2]. However, magnetic resonance imaging (MRI) provides better soft-tissue contrast, enhancing tumour and organ-at-risk delineation [3]. Integrating MRI into radiotherapy workflows can improve treatment accuracy but presents challenges, particularly the lack of electron density information necessary for dose planning [4].

To address this, synthetic CT (sCT) generation has become a potential solution [5]. By converting MRI data into sCT images, clinicians can take advantage of both imaging techniques without requiring extra CT scans, thereby reducing patient exposure to ionizing radiation and improving the workflow [5]. Recent advancements in deep learning, particularly generative adversarial networks (GANs), have facilitated significant progress in this area. For instance, Nie et al. (2017) [6] developed a context-aware GAN to estimate CT images from corresponding MR images, demonstrating the potential of data-driven approaches in this domain.

The SynthRAD2023 Challenge further emphasises the importance of sCT generation in radiotherapy. This initiative provided a comprehensive dataset of paired CT and MRI images from multiple centres, enabling the development and evaluation of image synthesis algorithms for radiotherapy purposes [7]. These collaborative efforts aim to compare different sCT generation methods, encouraging progress that could result in more accurate and efficient radiotherapy planning.

This study explores the use of deep learning models for synthetic CT (sCT) generation from MRI, focusing on Pix2Pix and Autoencoder architectures.

Additionally, CycleGAN+Pix2Pix was trained but ultimately not utilised in the final evaluation. Model performance was assessed using key quantitative metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise Ratio (SNR).

Methods

Deep Networks Overview

This study employs **Pix2Pix**, a conditional Generative Adversarial Network (GAN), and an **Autoencoder** to synthesise CT images from MR scans. **Pix2Pix** was chosen for its effectiveness in image-to-image translation, using adversarial training to generate high-quality CT images. The generator learns to map MR images to synthetic CTs, while the discriminator evaluates their realism. To maintain anatomical integrity, cycle consistency loss ensures that generated CT images can be mapped back to MR images. As a baseline model, an **Autoencoder** was implemented, offering a simpler, deterministic approach to reconstruct CT images. Using an encoder-decoder structure, the Autoencoder compresses MR image representations into a latent space and reconstructs them into CT images. This model acts as a comparison to the more complex generative approach.

Training Process

The **Pix2Pix** model was trained using paired MR-CT images, where the generator learns to synthesise CTs while the discriminator distinguishes real from generated images. Cycle consistency loss was applied to ensure that crucial MR image features were retained in the generated CTs. This adversarial setup was chosen for its ability to produce high-quality CT images, which are essential for radiotherapy planning. In contrast, the **Autoencoder** was trained using Mean Squared Error (MSE) loss, focusing on minimising pixel-wise differences between the generated and target CT images. The Autoencoder learns a direct mapping between MR and CT domains without adversarial supervision, serving as a simpler, deterministic alternative for comparison.

Hyperparameters and Loss Functions

For **Pix2Pix**, a learning rate of **0.001** with **cosine annealing** was used for gradual decay, and a **batch size of 4** was selected to balance memory efficiency and training stability. The model was optimised using three key loss functions: **adversarial loss** to ensure realism in CT images, **cycle consistency loss** to preserve MR anatomical features, and **L1 loss** to minimise pixel-level differences between generated and real CTs. The **Autoencoder**, trained with the **Adam optimizer** at a learning rate of **0.001**, used **MSE loss** to minimise reconstruction error. To stabilise training, batch normalization was applied to both models.

Architecture Changes and Modifications

The **Pix2Pix architecture** consists of an encoder-decoder framework with **batch normalization** and **LeakyReLU activations**. The decoder includes **upsampling layers** to refine the output, ensuring accurate CT reconstruction while keeping the anatomical structure intact. **Cycle consistency loss** further improves the quality of generated CT images by ensuring they can be reversed to MR images. The **Autoencoder architecture** follows a similar encoder-decoder structure but without adversarial components, focusing purely on reconstructing CT images.

Regularisation and Data Augmentation

Regularisation techniques were employed to make the model stronger- more robust and more adaptable. **Batch normalization** was applied in both models to prevent overfitting and ensure stable training. **Data augmentation** techniques, including **rotations, flips, and cropping**, were used to enhance the diversity of the training dataset. To prevent training instability, **gradient clipping** was applied in Pix2Pix.

Pre-training and Other Modifications

No **pre-training** was used for Pix2Pix; the model was trained directly on paired MR-CT datasets. The introduction of **cycle consistency loss** in Pix2Pix played a crucial role in ensuring the generated CT images could be mapped back to their respective MR images, improving anatomical accuracy.

Clinical Relevance

The main goal of these models is to help with radiotherapy planning by creating synthetic CT images from MR scans, which reduces the need for radiation exposure. The ability to create high-quality CT images from MR data improves treatment planning, especially for dose calculations and therapy guidance. To evaluate the performance and clinical applicability of these models, **Mean Absolute Error (MAE)**, **Peak Signal-to-Noise Ratio (PSNR)**, and **Structural Similarity Index (SSIM)** were used as quantitative assessment metrics. These measures ensure that the synthesised CT images are not only visually realistic but also clinically relevant for precise radiotherapy applications.

Experiments

PyTorch CycleGAN and Pix2Pix

For this experiment, I prepared a dataset of MRI and CT images to train a Pix2Pix model for MRI-to-CT translation using the PyTorch CycleGAN and Pix2Pix framework. First, I preprocessed the data by converting NIfTI (.nii.gz) files into PNG

format. This involved normalising the image intensities to a 0–255 range and extracting individual slices from the volumetric data. The dataset was then split into training (80%) and validation (20%) subsets, ensuring a balanced distribution of patient data. The preprocessed images were organised into appropriate directories following the structure required by the Pix2Pix framework. Training was performed using the `train.py` script with the `--model pix2pix` option, directing the model to learn the mapping from MRI (domain A) to CT (domain B). The training was executed on a CPU (`--gpu_ids -1`), allowing the model to optimise its parameters over multiple epochs. This approach aimed to create realistic CT images from MRI scans, making medical image translation easier.

Autoencoder Evaluation

The autoencoder was implemented to learn a mapping between MRI and CT images using a supervised approach. The model consists of an encoder-decoder structure, where the encoder extracts features from MRI images and the decoder reconstructs corresponding CT images. It was chosen as a baseline due to its simplicity and efficiency in medical imaging tasks.

Clinically, the autoencoder offers a straightforward comparison against more complex models like Pix2Pix, providing a solid baseline for image generation. Technically, while it doesn't generate new data from noise like GANs, it effectively learns meaningful features from MRI images to reconstruct plausible CT images. The model was evaluated using standard metrics like PSNR, SSIM, and MAE, ensuring a strong comparison with the other method. Data preprocessing, such as volume normalization and resizing, was used to improve model training and performance.

Pix2Pix Evaluation

To evaluate the performance of the Pix2Pix model in generating synthetic CT images from MRI, a systematic evaluation pipeline was implemented. The evaluation framework checks the model's outputs across multiple training epochs, calculating key quantitative metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise Ratio (SNR). The evaluation script loads Pix2Pix-generated CT images alongside real CT images and MRI inputs, normalizes them, and calculates differences in intensity and structure.

Additionally, gradient-based metrics, such as gradient error and normalized cross-correlation (NCC), were included to assess edge preservation and image fidelity. The results are visualised with comparison images and plotted metric trends over training epochs, giving an understanding of how the model performs and improves. This automated evaluation process ensures results can be repeated and allows for easy comparison between different models.

Results

In this section, there is a comprehensive comparison of the performance of the Pix2Pix and Autoencoder models in the context of CT image reconstruction. The results highlight key differences in model accuracy, image quality, and signal integrity, emphasising the strengths and limitations of each approach for clinical applications such as radiotherapy planning.

Model	AVG. MAE	AVG. MSE	AVG. PSNR	AVG. SSIM	AVG. SNR
Pix2Pix	0.0748	0.0113	20.51	0.4882	14.47
Autoencoder	0.1228	0.0507	13.58	0.5393	5.42

Table 1. Quantitative Comparison of Pix2Pix and Autoencoder Model Performance on CT Image Reconstruction

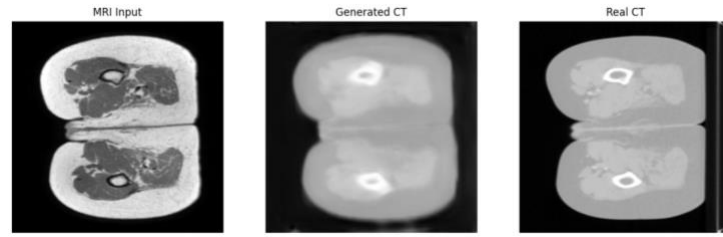


Fig. 1. Autoencoder Evaluation Results- Patient 1PA020, Slice 20

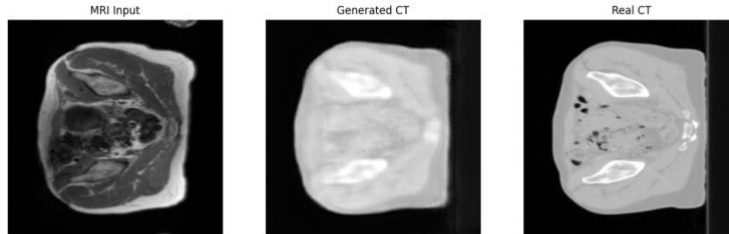


Fig. 2. Pix2Pix Evaluation Results- Epoch 200 (Final epoch)

The experimental results show clear performance differences between the Pix2Pix GAN and the Autoencoder models. Quantitatively, Pix2Pix outperformed the Autoencoder in most error metrics, achieving a lower MAE (0.0748 vs. 0.1228) and MSE (0.0113 vs. 0.0507), indicating reduced reconstruction error. The significantly higher PSNR score (20.51 vs. 13.58) suggests that Pix2Pix produces images with less noise and better accuracy to the ground truth. Interestingly, despite its higher error rates, the Autoencoder achieved a slightly better SSIM (0.5393 vs. 0.4882), suggesting better preservation of structure, though it may reduce overall accuracy.

Signal quality is also very important, especially in radiotherapy, where clearly defining tissue boundaries is crucial. Pix2Pix demonstrated a substantially higher SNR (14.47 vs. 5.42), indicating clearer signal distinction and reduced noise interference. This is an especially important finding, since higher SNR is essential for accurately defining anatomical structures in CT synthesis for radiotherapy planning.

A qualitative assessment of the generated images supports these findings. Pix2Pix produced sharper images with more distinct contrast between different tissue types, leading to better preservation of fine anatomical details and clearer boundary definition. In contrast, the Autoencoder-generated images appeared blurrier, with less distinct boundaries and smoothed-out details. While the Autoencoder maintained general structural relationships, losing fine details could affect its reliability for clinical use where precise anatomical definition is needed.

Discussion

From a clinical perspective, the superior image quality and lower error rates of Pix2Pix suggest that it would be more suitable for tasks that need high spatial accuracy, like radiotherapy planning. The sharper images and higher SNR help with more accurate target volume definition and dose calculations. However, both models achieved relatively low SSIM scores (< 0.6), showing that there is still work to be done in preserving structure before they can be used in clinical settings.

In terms of practical implementation, Pix2Pix also demonstrated a significant advantage in computational efficiency, completing 200 training epochs in just 12 hours, compared to the Autoencoder, which required 55 hours for 50 epochs. This difference has important implications for model retraining, adaptation to new datasets, and integration into clinical workflows. Both models were trained on a MacBook 2015 Intel version, highlighting their accessibility, but Pix2Pix's faster training time makes it a more practical choice for real-world applications.

Conclusion

Despite these advantages, there is scope for further improvement. Both models could improve in structural similarity, possibly by combining the Autoencoder's strength in preserving structure with Pix2Pix's ability to generate details. Further validation on a variety of patient groups and across different anatomical regions is needed to ensure reliability. Also, improving the designs with loss functions tailored for clinical use and using advanced data augmentation methods could make the models more adaptable.

Overall, Pix2Pix appears to be the more suitable model for CT synthesis in radiotherapy planning, offering better image quality, lower error rates, and faster training. However, more testing and improvements in structural similarity are needed before it can be used in clinical settings.

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